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ALICE L WALTON SCHOOL OF MEDICINE

Nonbacterial Thrombotic Endocarditis

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Adapted from: Nonbacterial Thrombotic Endocarditis

Listen to this Brick
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Learning Objectives (4)
Versions

After completing this brick, you will be able to:

1. Differentiate the clinical presentation of nonbacterial thrombotic endocarditis (NBTE).
2. Explain the pathophysiology of NBTE.
3. Interpret clinical findings to properly diagnose NBTE.
4. Outline the management plan of NBTE.

What condition underlies most cases of NBTE?

How Do Patients with Nonbacterial Thrombotic Endocarditis Present? (LO #1)

Patients with NBTE are often asymptomatic. In fact, it is most often diagnosed at autopsy, detected as an incidental finding. There may be no indications of valvular or heart dysfunction. This is because since the valvular lesions are non-infectious, they rarely result in destruction of the valve, as commonly occurs in infective endocarditis.

When present, symptoms are typically because of **thromboembolism** rather than due to direct valvular damage. In some cases, patients may not be diagnosed until after emboli have occurred. Common sites of embolization include the spleen, kidney, skin, and extremities that may present as flank pain, hematuria, rash, and the central nervous

Systemic Lupus Erythematosus

Systemic lupus erythematosus (SLE) is a multisystem autoimmune disease that can show symptoms in many organ systems, including the heart (pericarditis, or less often NBTE). The mechanism of vegetation formation is not well defined. Immune complex formation has been implicated, as the vegetations have been shown to contain immune complexes and cellular debris.

> CLINICAL CORRELATION
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> Some pathologists separate the sterile endocarditis of SLE into a separate diagnosis termed **Libman-Sacks endocarditis**, but this is not uniformly agreed upon. Others use the eponym (Libman-Sacks endocarditis) for a form of NBTE with very large thrombi.

Antiphospholipid Syndrome

Antiphospholipid syndrome (APS) occurs when autoantibodies attack cell membranes, causing excess clot formation. APS can occur as an idiopathic disorder but also in association with SLE or infections.

- Vegetations in NBTE are most often small, sterile, and nondestructive in contrast with the destructive vegetations of infective endocarditis.
- Vegetations in NBTE may embolize to different parts of the body.
- Diagnosis is made by echocardiography.
- Blood cultures negative for infective endocarditis support the diagnosis.
- Treatment is of the underlying cause (eg, SLE, APS, malignancy).
- Anticoagulation should be used if no contraindication exists.

Review Questions

1. Which of the following mechanisms most commonly leads to nonbacterial thrombotic endocarditis?

A. ○ Autoimmune response against the pericardium
B. ○ Colonization of the heart valve by *Staphylococcus aureus*
C. ● Hypercoagulable state
D. ○ Malignant cancer metastasizing to the heart valves
E. ○ Structural valvular disease

Hide Explanation

The correct answer is hypercoagulable state (C). An underlying hypercoagulable state, leading to wide most patients with nonbacterial thrombotic endocarditis (NBTE). An autoimmune response against the pericardium (A) sometimes occurs after a myocardial infarction but is not a part of the pathophysiology of NBTE. Colonization of the heart valve by *Staphylococcus aureus* (B) is incorrect because there is no infectious component in NBTE. A malignant cancer metastasizing to the heart valves (D) does not cause the formation of vegetations. Underlying structural valvular disease (E) is not a key cause of NBTE.

2. Which of the following is a common predisposing factor associated with nonbacterial thrombotic endocarditis?

A. ○ IV drug use
B. ● Malignancy
C. ○ Polyarteritis nodosa
D. ○ Progressive systemic sclerosis
E. ○ Rheumatic heart disease

Hide Explanation

The correct answer is malignancy (B). Malignancies can produce procoagulant proteins that cause a hypercoagulable state. The underlying hypercoagulable state is implicated in the pathophysiology of nonbacterial thrombotic endocarditis (NBTE). IV drug use (A) is the most common cause of tricuspid infective endocarditis. Polyarteritis nodosa (ocarditis. Progressive systemic sclerosis (D) umatic heart disease (E) may cause valvular damage and can set up the scene for the formation of an infectious vegetation.

3. A patient previously diagnosed with systemic lupus erythematosus comes to the clinic with history of sudden-onset weakness of her right leg. A murmur is auscultated on cardiac examination that was not documented in previous visits. What is the most likely complication that this patient may have developed?

A. ○ Acute infective endocarditis
B. ● Brain embolization
C. ○ Cardiomyopathy
D. ○ Septic shock
E. ○ Valvular degeneration

Hide Explanation

The correct answer is brain embolization (B). This patient has systemic lupus erythematosus, a predisposing condition to nonbacterial thrombotic endocarditis (NBTE). This, associated with the presence of a heart murmur, signifies that she has NBTE. Since the vegetations of NBTE are only loosely adherent to the valves, they are more prone to det at risk for acute infective endocarditis cess. Cardiomyopathy (C) is a possible complication but is not emergently debilitating. Septic shock (D) is also not a high risk for this patient. Like cardiomyopathy, valvular degeneration (E) will take years to develop the symptoms of heart failure.

4. What is the best clinical test to diagnose valvular vegetations?

A. ○ Anti-neutrophil cytoplasmic antibodies (ANCAs)
B. ○ Blood cultures
C. ○ Chest x-ray
D. ○ D-dimers
E. ● Echocardiogram

Hide Explanation

The correct answer is echocardiogram (E). Patients presenting with a new heart murmur or signs and symptoms of new-onset heart failure should get an echocardiogram, as it is the best test to detect valvular vegetations. Anti-neutrophil cytoplasmic antibodies (A) can be useful in the evaluation of certain autoimmune conditions, but do not have utility in identifying cardiac valvular lesions. Blood cultures (B) are a key component in the workup of endocarditis, but may be positive (infective endocarditis) or negative (culture negative infective endocarditis or NBTE), and as such do not assist with positively identifying valvular vegetations. Valvular vegetations are not re ay (C). D-dimers (D) are a useful laboratory , but do not assist with diagnosing valvular vegetations.

5. The management of NBTE involves treatment meant to prevent which known complication?

A. ○ Diastolic heart failure
B. ○ Valve destruction
C. ● Embolization
D. ○ Myocarditis
E. ○ Pleural effusion

Hide Explanation

The correct answer is embolization (C). Systemic embolization is the most common and clinically significant complication of NBTE. One of the most common places for embolization is the brain (manifesting as stroke), but the heart, kidney, and spleen are also commonly affected. Unless a contraindication exists, patients with NBTE should be treated with anticoagulants to prevent embolization. Importantly, the vegetations of NBTE are nondestructive, which is in contrast to the valve destruction (B) characteristic of the vegetations associated with infective endocarditis. Diastolic heart failure (A), myocarditis (D), and pleural effusions are not common or significant complications associated with NBTE.

How would you rate this Brick?

References

- https://heart.bmj.com/content/108/20/1637.abstract
- https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7100943/
- https://www.sciencedirect.com/science/article/abs/pii/S0033062022001086
- https://www.uptodate.com/contents/nonbacterial-thrombotic-endocarditis?search=nonbacterial%20thrombotic%20endocarditis&source=search_result&selectedTitle=1~28&usage_type=default&display_rank=1
- Kumar V, Abbas AK, Aster JC, eds. *Robbins Basic Pathology*. 10th ed. Elsevier; 2018.

Review Learning Objectives
Check off the objectives when you feel comfortable with the concept

- https://www.sciencedirect.com/science/article/abs/pii/S0033062022001086
- https://www.uptodate.com/contents/nonbacterial-thrombotic-endocarditis?search=nonbacterial%20thrombotic%20endocarditis&source=search_result&selectedTitle=1~28&usage_type=default&display_rank=1
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Objective Confidence: 4/4